

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE – GRADE 10 | SUB-STRAND 1.7: MICROORGANISMS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have spent 8 lessons investigating the invisible world of microorganisms. Now it is time to bring everything together. Your task is to write a Final Explanation that answers the driving question fully and clearly, using evidence and scientific vocabulary you have learned throughout this unit.

DRIVING QUESTION: Why did the ugali and chapati grow fuzzy patches overnight — and how can understanding the invisible organisms responsible help us protect our health and our food in Kenya?

READ THIS BEFORE YOU BEGIN:

- Use scientific vocabulary from the unit (e.g., fungi, bacteria, spores, decomposition, germination, nutrients, reproduction, pathogen, contamination, fermentation).
- Support every claim you make with evidence — from your experiments, observations, readings, videos, or class discussions.
- Connect your explanation back to the Kisumu kitchen phenomenon: the uncovered sufuria of ugali and the half-eaten chapati left on the counter overnight.
- Your explanation should be thorough enough that someone who was NOT in your class could read it and understand both what happened in the sufuria and why it matters for health and food safety in Kenya.
- Write in full sentences and organise your response section by section using the prompts below.
- Aim for approximately 600–900 words across all sections combined.

Section 1: What Were the Fuzzy Patches? (Identifying the Microorganisms)

Identify and describe the microorganisms responsible for the fuzzy patches on the ugali and chapati. In your response:

- Name at least TWO types of microorganisms that could cause the patches observed (white, green, and black fuzzy growth; sour

The fuzzy patches on the ugali and chapati were caused by microorganisms — living things so tiny they cannot be seen with the naked eye. The white, green, and black fuzzy growth is most likely mould, a type of fungus. Different colours indicate different species of mould: for example, *Rhizopus* (white/grey, common on starchy foods like ugali) and *Aspergillus* (green or black, common on grain-based foods like chapati). The sour smell and slimy areas are likely caused by bacteria, such as *Lactobacillus* or other decomposing bacteria, which break down the food chemically and release acidic waste products. Fungi are eukaryotic microorganisms — their cells have a nucleus — while bacteria are prokaryotic, with no membrane-bound nucleus. Both can reproduce very rapidly under the right conditions. The variation in colour and texture across the same food surface is because several species of microorganisms can colonise the same food at the same time, each growing in slightly different areas depending on moisture, nutrients, and competition.

smell; slimy texture). – Explain what microorganisms ARE — their key characteristics (size, cell type, ability to reproduce, etc.). – Explain why different patches were different colours and textures. – Use specific scientific vocabulary in your answer.	
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Section 2: Where Did They Come From? (Spores, Spread, and Conditions for Growth)

Explain how the microorganisms got onto the ugali and chapati and why they grew so quickly overnight. In your response: – Describe how mould spores and bacteria are spread (air, surfaces, hands, etc.). – Explain the conditions that allowed rapid growth overnight in the Kisumu kitchen (warmth, moisture, food as a nutrient source, uncovered container). – Reference what you learned about the life cycle of fungi, including spore germination. – Explain why the food did NOT need to be touched by anyone for microorganisms to appear.	Many people are surprised that the ugali and chapati grew mould even though no one touched them — but microorganisms do not need to be carried by hands or insects. Mould reproduces by releasing microscopic spores into the air. These spores are invisible, extremely light, and float freely in the environment, including inside homes in Kisumu. When spores land on a suitable surface — warm, moist food with nutrients — they germinate and grow into new mould colonies, producing the fuzzy structures we can see. The Kisumu kitchen provided ideal conditions: the ugali and chapati are rich in carbohydrates (starch and sugars), which serve as food for microorganisms. The kitchen was warm (Kenya's warm climate accelerates microbial growth), and the food was uncovered, exposing it directly to airborne spores and bacteria. Moisture from the cooked ugali also created a humid microenvironment on the food surface. Bacteria, which reproduce by binary fission, can double in number every 20 minutes under warm conditions — meaning millions of bacteria could have colonised the food within just a few hours. Together, these factors explain why the patches appeared as quickly as overnight.
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Section 3: Are Microorganisms Always Harmful? (Roles of Microorganisms in Daily Life)

Explain that microorganisms are not always harmful — they play important roles in	While the mould on the ugali and chapati is unwanted and potentially harmful, microorganisms are not always our enemies. Many microorganisms play vital roles in Kenyan daily life. For example, Lactobacillus bacteria are used in the fermentation of mursik (a traditional Kalenjin fermented milk), giving it its distinctive sour taste and making it safe to store. Yeast (<i>Saccharomyces cerevisiae</i>)
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everyday life in Kenya. In your response: – Give at least TWO examples of beneficial microorganisms and explain what they do (you may use examples from food, agriculture, medicine, or the environment). – Give at least TWO examples of harmful microorganisms and the diseases or problems they cause in Kenya. – Explain the concept of decomposition and why it is important for the environment, using a Kenyan context.	is a fungus used in baking mandazi and bread — it ferments sugars and releases carbon dioxide gas, making dough rise. In agriculture, nitrogen-fixing bacteria such as <i>Rhizobium</i> live in the root nodules of bean plants (common in farms across the Rift Valley) and convert atmospheric nitrogen into a form plants can absorb, acting as a natural fertiliser. However, some microorganisms are pathogens — they cause disease. <i>Salmonella</i> bacteria can contaminate poorly stored food like cooked ugali and cause typhoid and food poisoning, which is a real public health concern in Kenya. <i>Plasmodium</i>, a protist spread by <i>Anopheles</i> mosquitoes, causes malaria, one of Kenya's leading causes of illness. Fungi like <i>Candida</i> can cause infections in humans with weakened immune systems. Microorganisms also perform the essential ecological role of decomposition: bacteria and fungi break down dead organic matter — including food scraps, fallen leaves, and animal waste — into simpler substances that enrich the soil. Without decomposers, nutrients would not be recycled, and farmland across Kenya would lose its fertility.
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Section 4: How Can We Protect Our Food and Health? (Prevention and Food Safety in Kenya)

Apply your understanding of microorganisms to explain specific, evidence-based strategies for protecting food and health in Kenya. In your response: – Describe at least THREE practical food preservation or food safety methods relevant to Kenyan households, and explain the science behind WHY each method works (link each method to what it does to microorganism growth). – Discuss at least ONE personal hygiene or public health practice that reduces the spread of harmful microorganisms. – Where possible, refer to specific Kenyan contexts	Understanding how microorganisms grow helps us make smarter decisions about food safety. Since microorganisms need warmth, moisture, and nutrients to thrive, we can protect food by removing or reducing one or more of these conditions. First, refrigeration slows microbial growth by lowering the temperature below the optimal range for most bacteria and fungi — Kenyan households and school canteens in Nairobi that have access to refrigerators should store cooked ugali and chapati promptly after meals. Second, covering food — even with a simple lid or cloth — reduces exposure to airborne spores and insects that carry bacteria, making it one of the easiest and most affordable food safety steps for households in Kisumu or Mombasa. Third, drying and sun-drying is a traditional and effective preservation method: removing moisture prevents microbial growth, which is why dried maize stored after harvest in the Rift Valley lasts much longer than fresh maize. Salting and fermentation (as in fermented vegetables or mursik) create an environment that is too acidic or too salty for harmful microorganisms to survive, while encouraging beneficial ones. For personal hygiene, regular handwashing with soap and water before handling food removes bacteria and fungal spores from the skin. In public health, campaigns by Kenya's Ministry of Health that promote proper food handling at markets — such as covering meat at Kisumu's open-air markets — directly apply microbiology to reduce the spread of foodborne pathogens like <i>Salmonella</i> and <i>E. coli</i>.
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(e.g., markets in Kisumu, food stored during harvest season in the Rift Valley, school canteens in Nairobi).	
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Section 5: Revised Explanation — Answering the Driving Question

Now write a final, concise paragraph (8–12 sentences) that directly answers the driving question. Think of this as your 'scientist's conclusion.' In your response: – State clearly what caused the fuzzy patches on the ugali and chapati. – Explain the mechanism (how they got there, why overnight, what conditions helped). – Connect this to the bigger picture: what does understanding these invisible organisms mean for health and food safety in Kenya? – Reflect briefly on how your thinking has changed since Lesson 1, when you first saw the photographs of the spoiled food.	The fuzzy white, green, and black patches on the ugali and chapati were caused by microorganisms — specifically moulds (fungi such as <i>Rhizopus</i> and <i>Aspergillus</i>) and bacteria — that were present in the air of the Kisumu kitchen as invisible spores and cells. When the food was left uncovered overnight, these spores landed on the warm, moist, nutrient-rich surface and rapidly germinated and reproduced, forming the visible colonies we saw by morning. No one needed to touch the food and no insects were required — the microorganisms arrived through the air and found perfect conditions to grow. The warm Kenyan climate, the moisture in freshly cooked ugali, and the starch-rich nutrients in both foods accelerated their growth dramatically. This phenomenon is not unique to one kitchen — it is happening constantly in homes, markets, and schools across Kenya, and it carries real consequences for public health. Foodborne illnesses caused by bacteria like <i>Salmonella</i> and <i>E. coli</i> affect thousands of Kenyans each year, particularly in areas where food storage infrastructure is limited. However, understanding the biology of microorganisms gives us the tools to fight back: covering food, refrigerating leftovers, drying and fermenting produce, and practising good hygiene are all science-backed strategies that interrupt microbial growth. Microorganisms are not simply enemies — they are also partners in food production, soil health, and medicine. At the start of this unit, the fuzzy patches seemed mysterious and even frightening. Now, they are explainable: they are the visible evidence of an invisible living world that has always surrounded us, and understanding that world is one of the most powerful things science can offer us.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific Accuracy & Use of Vocabulary	Correctly identifies and names specific microorganisms (e.g., <i>Rhizopus</i> , <i>Aspergillus</i> , <i>Lactobacillus</i> , <i>Salmonella</i>). Uses 8 or more unit vocabulary terms accurately and in context throughout the explanation.	Correctly identifies types of microorganisms (fungi, bacteria) with at least one specific name. Uses 5–7 unit vocabulary terms accurately.	Identifies microorganisms in general terms only (e.g., 'germs' or 'mould') with few or no specific names. Uses fewer than 5 unit vocabulary terms, or uses some terms inaccurately.

Explaining the Phenomenon (Kisumu Kitchen)	Provides a complete, mechanistic explanation of how spores arrived on the food, why conditions in the Kisumu kitchen supported rapid overnight growth, and why different patches had different colours and textures. Explanation is logically sequenced.	Explains how microorganisms reached the food and names at least two growth conditions (e.g., warmth, moisture). Mentions spore germination or binary fission. May lack full mechanistic detail on colour/texture variation.	Provides a partial or vague explanation of the phenomenon. May state that microorganisms 'were already there' without explaining spore dispersal or growth conditions. Little or no connection to the specific Kisumu context.
Breadth of Microorganism Roles (Beneficial & Harmful)	Clearly distinguishes between beneficial and harmful microorganisms with at least two specific examples of each. Examples are relevant to Kenya (e.g., mursik fermentation, malaria, nitrogen fixation, food poisoning). Decomposition is explained with ecological significance.	Gives at least one beneficial and one harmful example with correct explanations. At least one example uses a Kenyan context. Decomposition is mentioned but may lack ecological depth.	Lists examples of harmful microorganisms only, or beneficial examples are vague (e.g., 'some bacteria are good'). Little or no Kenyan context. Decomposition is absent or incorrectly described.
Application to Food Safety & Health in Kenya	Describes three or more food safety or preservation strategies with clear scientific reasoning for why each works (linking directly to disrupting microbial growth conditions). At least two strategies are grounded in specific Kenyan household, market, or school contexts.	Describes two or more food safety strategies with some scientific reasoning. At least one Kenyan context is referenced. The link between strategy and microbial biology is mostly clear.	Lists one or two food safety tips without explaining the science behind them (e.g., 'cover your food' with no explanation of why). Little or no Kenyan context. Strategies feel copied rather than understood.
Quality of Final Driving Question Response	The concluding paragraph directly and fully answers the driving question in 8–12 sentences. It synthesises all unit learning (organism identity, mechanism, conditions, broader health implications), reflects on change in thinking, and communicates clearly for a non-specialist reader. Writing is confident and well-structured.	The concluding paragraph answers the driving question with most key ideas addressed. Some synthesis is present, and there is a brief reflection on prior thinking. Writing is mostly clear with minor gaps in structure or depth.	The concluding paragraph is incomplete, too brief, or does not directly answer the driving question. Key ideas from the unit are missing or only superficially mentioned. Little or no reflection on how understanding has changed.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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